Project Initialization and Planning Phase

Date	10 july 2026
Team ID	
Project Title	EV Data Analytics & Visualization Dashboard
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	Develop an **EV Data Analytics & Visualization Dashboard** that analyzes EV charging, battery performance, and vehicle efficiency using Tableau to support better decision-making.
Scope	The project collects EV datasets, stores them in a MySQL database, preprocesses the data, and creates interactive Tableau dashboards. Users can analyze charging patterns, battery health, driving range, and compare EV models through filters, KPIs, and reports.
Problem Statement	
Description	The increasing adoption of Electric Vehicles generates large volumes of charging and battery data. Without a centralized analytics platform, it is difficult for city planners, fleet managers, and customers to monitor charging demand, evaluate battery performance, and compare EV models effectively.
Impact	The proposed solution provides interactive dashboards that visualize EV charging behavior, battery efficiency, and vehicle performance. This helps optimize charging infrastructure, improve fleet operations, reduce operational costs, and support sustainable transportation planning.
Proposed Solution	
Approach	The system imports EV datasets into a MySQL database, validates and preprocesses the data using Python, and connects Tableau to generate interactive dashboards and reports. The dashboards provide insights into charging patterns, battery performance, and EV model comparisons with filters and KPIs for data-driven decision-making.

Key Features	• Interactive Tableau Dashboards • EV Charging Pattern Analysis • Battery Performance & Driving Range Analysis • Comparative EV Model Efficiency Dashboard • KPI Cards and Calculated Metrics • Interactive Filters and Drill-down Analysis • Dashboard & Story Reports • Flask Web Integration with Embedded Tableau Dashboards